HOSPITAL.FR

Network Infrastructure
Proof of Concept

ALPHA BLUEPRINT

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Complete Project Guide — From First VM to Full Production

Guide Complet du Projet — Du Premier VM à la Production Complète

[LEARN] All IT Levels: Beginner → Expert | Tous Niveaux IT : Débutant → Expert
[EN] [FR] Fully Bilingual English / Français | Entièrement Bilingue
[HOSPITAL] Domain: hospital.fr | 6 VMs | 4 VLANs | 8 GB Host RAM
[COURSE] Course-Ready Documentation | Documentation Prête pour les Cours IT

Version 3.0 — Production Ready | VirtualBox + pfSense + Windows Server 2022 + Ubuntu 22.04

0. How to Use This Guide / Comment Utiliser ce Guide

[EN] ENGLISH

This document is the Alpha Blueprint for the Hospital.fr Network POC. It is designed to work for three different audiences simultaneously. Before diving into Phase 1, read this section carefully to understand how to navigate the guide and get the most out of it.

0.1 Three Reader Tracks / Trois Parcours de Lecture

Track / Parcours	Who it is for / Pour qui	How to use it / Comment l'utiliser
[BEGINNER] BEGINNER Débutant	First time touching networking, VMs, or servers. You may have heard of IP addresses but never configured one.	Read every paragraph including the [LEARN] LEARN WHY callouts. They explain concepts in plain language. Never skip a step.
[INTERMEDIATE] INTERMEDIATE Intermédiaire	You know what an IP address is, have used Linux, and have set up a VM before — but have not done VLANs or AD.	Read all headings and steps. You can skip [LEARN] LEARN WHY callouts you already understand. Focus on tables and code blocks.
[CRITICAL] EXPERT Expert	Senior sysadmin / network engineer who wants the reference. You will build on this and adapt it for production or teach it.	Use as a technical reference. Skip to the phase you need. All credentials, IPs, and commands are in the tables.

0.2 Document Conventions / Conventions du Document

Symbol	Meaning / Signification
ℹ INFO	Background information, context, or optional reading
⚠ WARNING	Do this wrong and you will waste an hour. Read carefully.
[CRITICAL] CRITICAL	The single most common failure point. Do not skip.
✅ SUCCESS CHECK	Run this test before moving to the next phase.
[TIP] TIP	A professional shortcut or best-practice hint.
[LEARN] LEARN WHY	Explains the concept behind the step — for beginners and IT course learners.
Code block (blue bar)	A command to type exactly as shown — Courier New font, blue left border.
# Comment (green)	Not a command — explains what the command does. Do not type these.
[EN] ENGLISH / [FR] FRANÇAIS	Language-specific paragraph. Same content, different language.

0.3 Project Philosophy / Philosophie du Projet

[EN] ENGLISH

Every step in this guide answers three questions:

- WHY are we doing this? — understanding intent makes you a better engineer, not just a button-presser.
- WHAT is the goal? — knowing the destination helps you verify you got there.
- WHAT is the expected outcome? — you can always check if the step worked correctly.

[FR] FRANÇAIS

Chaque étape de ce guide répond à trois questions :

- POURQUOI faisons-nous cela ? — comprendre l'intention fait de vous un meilleur ingénieur.
- QUEL est l'objectif ? — connaître la destination vous aide à vérifier que vous y êtes arrivé.
- QUEL est le résultat attendu ? — vous pouvez toujours vérifier si l'étape a fonctionné.

0.4 Critical Rules Before You Start / Règles Essentielles Avant de Commencer

[CRITICAL] CRITICAL / CRITIQUE

RULE 1: Always start VMs in order: FW01 → DC01 → APP01 → MED01 → CLIENT01. Out-of-order boot = connectivity failure.

RULE 2: Set Promiscuous Mode to 'Allow All' on EVERY VM network adapter — without this, VLANs will NOT work in VirtualBox.

RULE 3: Take a VirtualBox snapshot before every major phase. It is your undo button.

RULE 4: pfSense web UI is only accessible from inside a VM (not your Windows host). Use DC01 browser.

RULE 5: YAML (netplan) files are space-sensitive. Use exactly 2 spaces per indent level. NEVER use Tab.

RÈGLE 1: Toujours démarrer les VMs dans l'ordre: FW01 → DC01 → APP01 → MED01 → CLIENT01.

RÈGLE 2: Mode Promiscuité sur 'Allow All' sur TOUS les adaptateurs réseau de chaque VM.

RÈGLE 4: L'interface Web pfSense n'est accessible que depuis une VM — utilisez le navigateur de DC01.

1. Project Architecture / Architecture du Projet

1.1 Executive Summary / Résumé Exécutif

[EN] ENGLISH

This project builds a fully segmented, enterprise-grade hospital network entirely inside VirtualBox on a single Windows PC with 8 GB of RAM. It demonstrates every major concept an IT professional needs to know: virtualization, firewalls, VLANs, Active Directory, DNS, web application hosting, medical system deployment, network monitoring, and security policy enforcement.

When complete, this lab is indistinguishable — from a networking perspective — from a real hospital data centre. The techniques, tools, and architecture are production-grade.

[FR] FRANÇAIS

Ce projet construit un réseau hospitalier entièrement segmenté, de qualité entreprise, entièrement dans VirtualBox sur un seul PC Windows avec 8 Go de RAM. Il démontre chaque concept majeur qu'un professionnel IT doit connaître : virtualisation, pare-feux, VLANs, Active Directory, DNS, hébergement d'applications web, déploiement de systèmes médicaux, surveillance réseau et application des politiques de sécurité.

1.2 Complete VM Inventory / Inventaire Complet des VMs

VM Name	Role / Rôle	OS	RAM	Storage	VLAN	Final IP	Boot Order
FW01	Firewall / Router / DHCP	pfSense 2.7+	1 GB	10 GB	ALL	10.10.x.1	1st — Always first
DC01	Domain Controller / DNS	Windows Server 2022	2 GB	40 GB	VLAN 10	10.10.10.10	2nd — Wait 3 min
APP01	Web App Server	Ubuntu Server 22.04	1.5 GB	25 GB	VLAN 20	10.10.20.20	3rd
MED01	Medical Systems / PACS	Ubuntu Server 22.04	1 GB	20 GB	VLAN 20	10.10.20.21	4th
CLIENT01	Staff Workstation	Windows 11 Pro	1.5 GB	30 GB	VLAN 30	10.10.30.30	5th — when testing
GUEST01	IoT / Guest Access	Windows 11	1 GB	20 GB	VLAN 40	10.10.40.40	When needed
TOTAL	—	—	8 GB	145 GB	—	—	Max 4 VMs at once

⚠ WARNING / AVERTISSEMENT

Never run all 6 VMs simultaneously on an 8 GB host. Maximum 4 at once.
Priority: FW01 + DC01 + APP01 + MED01 for server testing. Swap in CLIENT01 for workstation demos.
Use 'Start Headless' for servers you don't need a console on: Right-click VM → Start → Start Headless.

1.3 Network Architecture Diagram / Schéma d'Architecture Réseau

```
                        [ INTERNET ]
                             |
                   ┌─────────────────┐
                   │  FW01 pfSense   │   WAN: NAT (DHCP from host)
                   │  2.7+           │   LAN: em1 → VLAN trunk
                   └─────────────────┘
                             |
        ┌────────────────────┼──────────────────────┐
        |                    |                      |              |
   ┌────┴────┐          ┌────┴────┐          ┌──────┴──────┐   ┌────┴────┐
   │ VLAN 10 │          │ VLAN 20 │          │ VLAN 30     │   │ VLAN 40 │
   │Management│         │Clinical │          │ Staff       │   │IoT/Guest│
   │10.10.10.0│         │10.10.20.0│         │10.10.30.0   │   │10.10.40.0│
   │   /24   │          │   /24   │          │   /24       │   │   /24   │
   └────┬────┘          └────┬────┘          └──────┬──────┘   └────┬────┘
        |                    |                      |              |
    [DC01]            [APP01][MED01]          [CLIENT01]      [GUEST01]
  10.10.10.10         .20    .21              10.10.30.30     10.10.40.40
 Win Srv 2022   Ubuntu Ubuntu                 Win 11 Pro      Win 11
```

1.4 VLAN Design Table / Tableau de Conception VLAN

VLAN ID	Name / Nom	Subnet	pfSense Gateway	DHCP Range	Purpose / Justification
10	Management / Gestion	10.10.10.0/24	10.10.10.1	10.10.10.100-200	IT Infrastructure, DC, pfSense admin — highest trust
20	Clinical / Clinique	10.10.20.0/24	10.10.20.1	10.10.20.100-200	Medical servers, patient data, PACS — isolated from staff
30	Staff / Personnel	10.10.30.0/24	10.10.30.1	10.10.30.100-200	Workstations, office systems — controlled access to VLAN20
40	IoT / Guest / Invités	10.10.40.0/24	10.10.40.1	10.10.40.100-200	Internet only — zero access to internal VLANs

[LEARN] LEARN WHY / APPRENDRE POURQUOI

WHY SEGMENT THE NETWORK?

In a flat network (no VLANs), every device can see every other device. One infected PC could reach every medical record.

VLANs act like physical walls inside the network. A guest on VLAN 40 cannot even see that VLAN 10 or 20 exist.

In real hospitals this is mandatory: HIPAA (USA), HDS (France), and GDPR all require isolation of patient data.

POURQUOI SEGMENTER LE RÉSEAU? Dans un réseau plat, chaque appareil peut voir tous les autres. Les VLANs créent des murs virtuels.

1.5 Service Catalog / Catalogue des Services

Service	VM	IP:Port	Technology	Purpose / Rôle
Firewall / Router	FW01	10.10.10.1:443	pfSense 2.7+	Network security, VLAN routing, DHCP, NAT
Active Directory	DC01	10.10.10.10:389	Windows Server 2022	User auth, DNS, Group Policy, domain services
DNS	DC01	10.10.10.10:53	Windows DNS	Name resolution for all VLANs
Hospital Web Portal	APP01	10.10.20.20:80	Apache + PHP + MySQL	Staff portal, patient data, appointments
PACS Medical Imaging	MED01	10.10.20.21:8042	Orthanc	DICOM image storage and viewing
System Monitoring	APP01	10.10.20.20/zabbix	Zabbix 6.0	Infrastructure health, alerts, metrics
Database	APP01	10.10.20.20:3306	MySQL 8.0	Hospital portal data storage

2. Prerequisites & Downloads / Prérequis et Téléchargements

2.1 Host Machine Requirements / Exigences de la Machine Hôte

Requirement	Minimum	Recommended	Your Setup
RAM / Mémoire	8 GB	16 GB	8 GB — manageable with care
CPU / Processeur	4 cores, Intel VT-x or AMD-V enabled	6+ cores	Enable virtualisation in BIOS
Storage / Stockage	150 GB free	250 GB SSD	Use dynamic disks to save space
OS / Système	Windows 10 64-bit	Windows 11	Windows 10/11 — confirmed
Network / Réseau	Any active adapter	Gigabit	For bridged adapter SSH access
Virtualization	VT-x/AMD-V in BIOS	—	Must be enabled — check Task Manager → Performance → CPU

[LEARN] LEARN WHY / APPRENDRE POURQUOI

WHAT IS VIRTUALIZATION? Your CPU has a feature called VT-x (Intel) or AMD-V (AMD) that allows it to run multiple operating systems at once.

VirtualBox uses this to create 'virtual machines' — each VM thinks it is a real physical computer.

Without VT-x/AMD-V enabled in BIOS, VMs run very slowly or not at all.

To check: open Task Manager on Windows → Performance tab → CPU → look for 'Virtualization: Enabled'.

QU'EST-CE QUE LA VIRTUALISATION? VT-x/AMD-V permet au CPU d'exécuter plusieurs OS simultanément.

2.2 Software Downloads / Téléchargements Logiciels

Software	Where to Download	File Type	Notes
Oracle VirtualBox 7.x	virtualbox.org → Downloads	EXE installer	Also download the Extension Pack — same page
VirtualBox Extension Pack	virtualbox.org → Downloads	VBOX-EXTPACK	Required for USB 2.0/3.0 and better performance
pfSense 2.7+ ISO	pfsense.org → Download	AMD64 ISO (~1 GB)	Choose: AMD64, DVD Image (ISO), latest CE
Windows Server 2022	Microsoft Evaluation Center	ISO (~5 GB)	Free 180-day evaluation — search 'Windows Server 2022 evaluation'
Ubuntu Server 22.04 LTS	ubuntu.com/download/server	ISO (~1.5 GB)	Choose 22.04 LTS (Long Term Support)
Windows 11 Pro	Microsoft Evaluation Center	ISO (~5 GB)	Or use existing licensed copy — 90-day eval available

[TIP] TIP / CONSEIL

Download ALL ISOs before starting. Create a folder C:\ISO\ and put everything there.

Rename files clearly: pfsense-2.7.iso, winserver2022.iso, ubuntu22.iso, win11.iso
Total download size: approximately 12-15 GB. Use a fast connection.
Téléchargez TOUS les ISOs avant de commencer. Créez un dossier C:\ISO\ et mettez-y tout.

2.3 VirtualBox Installation / Installation de VirtualBox

[EN] ENGLISH

Install VirtualBox on your Windows host before creating any VMs.

1. Double-click the VirtualBox installer (.exe) → click Next through all screens → Install
2. When prompted about network interfaces: click Yes (it briefly disconnects your internet — this is normal)
3. After VirtualBox installs, double-click the Extension Pack (.vbox-extpack) → click Install → Agree
4. Launch VirtualBox Manager — you should see an empty VM list

[FR] FRANÇAIS

Installez VirtualBox sur votre hôte Windows avant de créer des VMs.

5. Double-cliquez sur l'installateur VirtualBox → cliquez Suivant → Installer
6. Après l'installation, double-cliquez sur le Extension Pack → Installer → Accepter
7. Lancez VirtualBox Manager — vous devriez voir une liste de VMs vide

PHASE 1 — Create All Virtual Machines

Créer Toutes les Machines Virtuelles

3. Create All Virtual Machines / Créer Toutes les Machines Virtuelles

ℹ INFO / NOTE

WHY THIS PHASE: Before installing any operating system, we create all VM 'shells' in VirtualBox.
Setting up the hardware configuration correctly now prevents having to redo it later.
KEY RULE: Every VM (except FW01) uses Internal Network named exactly 'LabNet' for its main adapter.
POURQUOI CETTE PHASE: Nous créons d'abord les coquilles VM vides pour configurer le matériel virtuel.
RÈGLE CLÉ: Chaque VM (sauf FW01) utilise le Réseau Interne nommé exactement 'LabNet'.

3.1 FW01 — pfSense Firewall VM

[LEARN] LEARN WHY / APPRENDRE POURQUOI

WHAT IS A FIREWALL? A firewall is the gatekeeper between your network and the internet.
pfSense is a free, enterprise-grade firewall that also acts as a router, DHCP server, and VLAN switch.
FW01 gets TWO network adapters: one facing the internet (NAT), one facing all the VMs (Internal Network 'LabNet').
QU'EST-CE QU'UN PARE-FEU? pfSense est un pare-feu gratuit de qualité entreprise qui agit aussi comme routeur et switch VLAN.

In VirtualBox Manager: click New (top-left) and configure:

Setting	Value	Notes
Name	FW01	Exactly this — used throughout the guide
Machine Folder	C:\VMs\ (or your preferred path)	Create this folder first
Type	BSD	Not Linux — this is FreeBSD-based
Version	FreeBSD (64-bit)	—
RAM / Memory	1024 MB	1 GB is sufficient for pfSense
Hard Disk	Create a virtual hard disk now	—
Hard Disk Type	VDI (VirtualBox Disk Image)	Native format — best performance
Storage	Dynamically allocated	Saves real disk space
Size	10 GB	pfSense needs very little space

FW01 Network Adapters — CRITICAL

After creating the VM, go to FW01 → Settings → Network:

Adapter	Attached to	Name / Details	Purpose
Adapter 1	NAT	(automatic)	WAN — internet connection through your host
Adapter 2	Internal Network	Name: LabNet (type exactly)	LAN — connects to all other VMs

[CRITICAL] CRITICAL / CRITIQUE

On Adapter 2, expand 'Advanced' and set Promiscuous Mode to: Allow All
This is MANDATORY for VLAN traffic to pass. Without it, VMs cannot communicate across VLANs.
Sur l'Adaptateur 2, développez 'Avancé' et réglez le Mode Promiscuité sur : Allow All

3.2 DC01 — Domain Controller VM

[LEARN] LEARN WHY / APPRENDRE POURQUOI

WHAT IS A DOMAIN CONTROLLER? In a Windows network, the Domain Controller is the 'master database' of all users, computers, and passwords.
When a doctor logs into any workstation in the hospital, DC01 verifies their identity.
Active Directory (the service running on DC01) also provides DNS — the phone book that translates names like 'app01.hospital.fr' into IP addresses.
QU'EST-CE QU'UN CONTRÔLEUR DE DOMAINE? C'est la base de données centrale de tous les utilisateurs et mots de passe Windows.

Setting	Value	Notes
Name	DC01	Domain Controller
Type	Microsoft Windows	—
Version	Windows 2022 (64-bit)	Select this exactly
RAM	2048 MB	2 GB minimum for Windows Server
Hard Disk	VDI, Dynamically allocated	—
Size	40 GB	Windows Server needs space for updates and AD database

Network: Settings → Network → Adapter 1:

- Attached to: Internal Network
- Name: LabNet ← must match exactly
- Advanced → Promiscuous Mode: Allow All ← MANDATORY

3.3 APP01 — Application Server VM

Setting	Value	Notes
Name	APP01	Web/Application Server
Type	Linux	—

Setting	Value	Notes
Version	Ubuntu (64-bit)	—
RAM	1536 MB	1.5 GB — enough for Apache + MySQL
Hard Disk	VDI, Dynamically allocated, 25 GB	—

- Adapter 1: Internal Network, Name: LabNet, Promiscuous Mode: Allow All

3.4 MED01 — Medical Systems Server VM

Setting	Value	Notes
Name	MED01	Medical / PACS Server
Type	Linux	—
Version	Ubuntu (64-bit)	—
RAM	1024 MB	1 GB — sufficient for Orthanc
Hard Disk	VDI, Dynamically allocated, 20 GB	—

- Adapter 1: Internal Network, Name: LabNet, Promiscuous Mode: Allow All

3.5 CLIENT01 — Staff Workstation VM

Setting	Value	Notes
Name	CLIENT01	Staff Workstation
Type	Microsoft Windows	—
Version	Windows 11 (64-bit)	—
RAM	1536 MB	1.5 GB minimum for Windows 11
Hard Disk	VDI, Dynamically allocated, 30 GB	—

- Adapter 1: Internal Network, Name: LabNet, Promiscuous Mode: Allow All

3.6 GUEST01 — IoT / Guest VM

Setting	Value	Notes
Name	GUEST01	Guest / IoT / Visitor
Type	Microsoft Windows	—
Version	Windows 11 (64-bit)	—
RAM	1024 MB	1 GB — minimal guest access

Setting	Value	Notes
Hard Disk	VDI, Dynamically allocated, 20 GB	—

- Adapter 1: Internal Network, Name: LabNet, Promiscuous Mode: Allow All

3.7 Summary — All VM Network Settings / Résumé

VM	Adapter 1	Adapter 2	Promiscuous Mode (ALL adapters)
FW01 (pfSense)	NAT	Internal Network: LabNet	Allow All on BOTH adapters
DC01	Internal Network: LabNet	—	Allow All
APP01	Internal Network: LabNet	—	Allow All
MED01	Internal Network: LabNet	—	Allow All
CLIENT01	Internal Network: LabNet	—	Allow All
GUEST01	Internal Network: LabNet	—	Allow All

✓ SUCCESS CHECK / VÉRIFICATION

VERIFY BEFORE CONTINUING: All 6 VMs appear in VirtualBox Manager.
Every VM shows the correct RAM and disk in the details panel.
All VMs (except FW01 Adapter 1) are connected to Internal Network 'LabNet'.
Promiscuous Mode is set to Allow All on every adapter of every VM.

PHASE 2 — Install & Configure pfSense

Installer et Configurer pfSense

4. pfSense — Firewall, Router & VLAN Engine

ℹ INFO / NOTE

WHY: pfSense is the FIRST thing installed and the LAST thing shut down. Everything depends on it.

GOAL: Install pfSense, assign WAN/LAN interfaces, create 4 VLANs, set up DHCP for each, and add firewall rules.

OUTCOME: A working multi-VLAN router. All VMs can get IP addresses and communicate through pfSense.

4.1 Install pfSense

[LEARN] LEARN WHY / APPRENDRE POURQUOI

pfSense is a FreeBSD-based firewall/router OS. You install it like any other OS from an ISO.

Once installed, it has a command-line console AND a web GUI you access from a browser.

pfSense prend en charge les VLANs nativement — c'est l'une des raisons pour lesquelles nous l'utilisons.

8. Select FW01 in VirtualBox → click the Settings gear → Storage → under 'Controller: IDE', click the empty CD icon → click the small CD icon on the right → Choose a disk file → select your pfSense ISO
9. Click OK → click the green Start arrow to boot FW01
10. Wait for the pfSense boot menu — press Enter to accept default boot option
11. When installer appears: accept Copyright → Install pfSense
12. Keymap: Continue with default keymap → OK
13. Partitioning: Auto (ZFS) → Proceed with Installation → Stripe (no redundancy) → select the virtual disk with Space → OK
14. Wait for installation to complete (2-5 minutes) → Reboot
15. When prompted, remove the ISO: VirtualBox menu → Devices → Optical Drives → Remove disk from virtual drive → press Enter in console

4.2 Assign Interfaces at Console

After reboot, pfSense shows a text console menu. You will now assign which interface is WAN and which is LAN.

16. At the menu, press 1 (Assign Interfaces)
17. 'Should VLANs be set up now?' → type n → Enter (we do VLANs in the web UI later)
18. 'Enter the WAN interface name:' → type em0 → Enter
19. 'Enter the LAN interface name:' → type em1 → Enter
20. 'Enter the Optional 1 interface name:' → press Enter to skip (leave blank)
21. Confirm the assignments → type y → Enter

After assignment, pfSense automatically sets LAN IP to 192.168.1.1. You will see:

```
LAN (em1)   -> v4: 192.168.1.1/24
WAN (em0)   -> v4: DHCP (gets IP from your host via NAT)
```

✅ SUCCESS CHECK / VÉRIFICATION

pfSense console menu should show the interface assignment and LAN IP 192.168.1.1.
You should see 16 options numbered 0-15 in the console menu.
Do NOT close or shut down FW01 — it must stay running.

4.3 Set LAN IP — Console Option 2

ℹ INFO / NOTE

If the LAN shows 192.168.1.1 already, skip this step. If it shows a different IP, set it now.

22. Press 2 (Set interface(s) IP address)
23. Select LAN (em1)
24. Enter new LAN IPv4 address: 192.168.1.1
25. Enter subnet mask (CIDR): 24
26. For gateway: press Enter (LAN has no upstream gateway)
27. Configure IPv6? → press Enter to skip
28. Enable DHCP server? → type y → Enter
29. Start of DHCP range: 192.168.1.100
30. End of DHCP range: 192.168.1.200
31. HTTP as webConfigurator protocol? → type y (for now)

4.4 Access pfSense Web UI from DC01

[CRITICAL] CRITICAL / CRITIQUE

pfSense web UI is NOT accessible from your Windows host machine.
You MUST access it from inside a VM on the LabNet network.
Start DC01 → wait for Windows to fully boot → open Edge or Internet Explorer on DC01.
L'interface Web pfSense N'EST PAS accessible depuis votre PC hôte Windows.
Vous DEVEZ y accéder depuis une VM sur le réseau LabNet — ouvrez Edge sur DC01.

32. Start DC01 VM (we will do full OS install in Section 5 — for now, if DC01 is already installed, use it to access pfSense)
33. Open browser on DC01 → navigate to: http://192.168.1.1
34. Login: admin / pfsense
35. pfSense Setup Wizard appears → click Next through all steps → accept defaults
36. At the 'Set Admin WebGUI Password' step: you may keep 'pfsense' for the lab or set your own
37. Click Finish → you are now in the pfSense Dashboard

[TIP] TIP / CONSEIL

Bookmark http://192.168.1.1 in DC01's browser right now. After VLAN migration it becomes https://10.10.10.1
If you get an SSL warning (HTTPS): click Advanced → Proceed. pfSense uses a self-signed certificate.

4.5 Create VLANs on pfSense

[LEARN] LEARN WHY / APPRENDRE POURQUOI

WHAT IS A VLAN? A VLAN (Virtual LAN) is a way to divide one physical network into multiple isolated logical networks.

pfSense creates VLAN sub-interfaces on the LAN adapter (em1). Each VLAN gets a number (tag) and its own IP range.

Traffic is labeled with the VLAN tag as it passes through the network — only devices on the same VLAN can see each other by default.

QU'EST-CE QU'UN VLAN? Un VLAN divise un réseau physique en plusieurs réseaux logiques isolés.

Votre interface LAN confirmée est em1 — c'est ce que nous utiliserons pour TOUS les VLANs.

Navigate to: Interfaces → Assignments → VLANs tab → click + Add

VLAN #	Parent Interface	VLAN Tag	Description	After saving: you will see
1 of 4	em1 (your confirmed LAN)	10	Management	em1.10 in the list
2 of 4	em1	20	Clinical	em1.20 in the list
3 of 4	em1	30	Administrative	em1.30 in the list
4 of 4	em1	40	Guest_IoT	em1.40 in the list

Click Save after each VLAN. After all 4 are created, you should see 4 entries in the VLAN list.

4.6 Assign VLANs as Interfaces

Navigate to: Interfaces → Assignments (Interface Assignments tab)

38. In the 'Available network ports' dropdown at the bottom, select 'VLAN 10 on em1 (Management)' → click Add
39. Repeat: select 'VLAN 20 on em1 (Clinical)' → Add
40. Repeat: select 'VLAN 30 on em1 (Administrative)' → Add
41. Repeat: select 'VLAN 40 on em1 (Guest_IoT)' → Add
42. Click Save — you now have OPT1, OPT2, OPT3, OPT4 in the interface list

4.7 Configure Each VLAN Interface

For each new interface, click its name in the list and configure it. Click Save + Apply Changes after each one.

Click this interface	Enable?	Description to set	IPv4 Config Type	IPv4 Address	Save & Apply?
OPT1	☑Check Enable	MGMT_VLAN10	Static IPv4	10.10.10.1 / 24	Yes — Apply Changes
OPT2	☑Check Enable	CLINICAL_VLAN20	Static IPv4	10.10.20.1 / 24	Yes — Apply Changes
OPT3	☑Check Enable	ADMIN_VLAN30	Static IPv4	10.10.30.1 / 24	Yes — Apply Changes
OPT4	☑Check Enable	GUEST_VLAN40	Static IPv4	10.10.40.1 / 24	Yes — Apply Changes

⚠ WARNING / AVERTISSEMENT

The orange 'Apply Changes' banner must appear and you must click it after saving each interface.
If you skip Apply Changes, the interface configuration is saved but NOT active.
La bannière orange 'Apply Changes' doit apparaître — vous devez cliquer dessus après chaque interface.

4.8 Enable DHCP Server on Each VLAN

[LEARN] LEARN WHY / APPRENDRE POURQUOI

WHAT IS DHCP? DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses to devices.
Without DHCP, every device would need a manually configured static IP — impractical at scale.
pfSense runs a DHCP server for each VLAN, handing out IPs only within that VLAN's range.
QU'EST-CE QUE DHCP? Il attribue automatiquement des adresses IP aux appareils — sans configuration manuelle.

Navigate to: Services → DHCP Server → click each VLAN tab and configure:

Tab to click	Enable DHCP?	Range: From	Range: To	DNS Servers (in 'DNS Servers' field)
MGMT_VLAN10	☑Enable	10.10.10.100	10.10.10.200	10.10.10.10
CLINICAL_VLAN20	☑Enable	10.10.20.100	10.10.20.200	10.10.10.10
ADMIN_VLAN30	☑Enable	10.10.30.100	10.10.30.200	10.10.10.10
GUEST_VLAN40	☑Enable	10.10.40.100	10.10.40.200	8.8.8.8

Click Save at the bottom of each tab. DHCP is active immediately — no Apply Changes needed here.

4.9 Add Firewall Rules — Pass All for Initial Setup

ℹ INFO / NOTE

pfSense blocks ALL inter-VLAN traffic by default. We add permissive rules now for testing.
After the project is working, Section 9 replaces these with precise security rules.
TEMPORARILY allowing all traffic lets us verify connectivity before adding complexity.

Navigate to: Firewall → Rules → click the MGMT_VLAN10 tab → click the ↑ Add button (adds rule at top)

Field	Set to
Action	Pass
Interface	MGMT_VLAN10
Address Family	IPv4
Protocol	Any
Source	Any
Destination	Any
Description	TEMP Allow All - Testing Phase

Click Save → click Apply Changes. Repeat this for CLINICAL_VLAN20, ADMIN_VLAN30, and GUEST_VLAN40 tabs.

✓ SUCCESS CHECK / VÉRIFICATION

CHECK: Status → Interfaces — all VLAN interfaces should show Status: UP with correct 10.10.x.1 IPs.
CHECK: Status → DHCP Leases — as VMs connect, their leases will appear here.
VÉRIFICATION: Interfaces → toutes les interfaces VLAN doivent afficher Statut: UP.

PHASE 3 — Install & Configure DC01

Domaine, Active Directory & DNS

5. DC01 — Windows Server 2022 Domain Controller

ℹ INFO / NOTE

WHY: DC01 is the identity and DNS backbone of the entire hospital network.
GOAL: Install Windows Server 2022, promote to Domain Controller, configure DNS for all VLANs.
OUTCOME: hospital.fr domain is live. All VMs can authenticate users and resolve DNS names.

5.1 Install Windows Server 2022

43. Select DC01 → Settings → Storage → attach Windows Server 2022 ISO to the DVD drive
44. Start DC01 → it boots from the ISO
45. Language: select your language → Next → Install Now
46. Select edition: Windows Server 2022 Standard (Desktop Experience) — this gives a GUI
47. Accept license → Custom: Install Windows only (advanced)
48. Select the virtual hard disk → Next → installation begins (10-20 minutes)
49. After reboot: set Administrator password — use: P@ssw0rd (or your own — note it carefully)
50. Remove ISO: Devices → Optical Drives → Remove disk from virtual drive
51. Log in: press Ctrl+Alt+Delete in VM (VirtualBox menu: Input → Insert Ctrl+Alt+Del) → Administrator / P@ssw0rd

5.2 Set Static IP on DC01

[LEARN] LEARN WHY / APPRENDRE POURQUOI

WHY STATIC IP? The Domain Controller MUST have a fixed IP address that never changes.
If DC01 got its IP from DHCP, it might change after a reboot — and all other VMs would lose DNS.
Servers always use static IPs. Workstations and guest devices use DHCP.
POURQUOI UNE IP STATIQUE? Le contrôleur de domaine doit avoir une IP fixe — s'il change, tout le réseau perd le DNS.

On DC01, open Control Panel → Network and Sharing Center → Change adapter settings:

52. Right-click the Ethernet adapter → Properties
53. Double-click 'Internet Protocol Version 4 (TCP/IPv4)'
54. Select 'Use the following IP address' and enter:

Field	Value	Notes
IP Address	10.10.10.10	DC01 VLAN 10 address — permanent, never changes

Field	Value	Notes
Subnet Mask	255.255.255.0	Standard /24 — covers 10.10.10.0 to 10.10.10.255
Default Gateway	10.10.10.1	pfSense VLAN 10 interface — the door to other VLANs
Preferred DNS Server	10.10.10.10	DC01 itself — it IS the DNS server
Alternate DNS Server	8.8.8.8	Google — fallback for internet resolution

55. Click OK → Close → Close
56. Verify: open Command Prompt (Win+R → cmd) → type: ipconfig /all
57. Confirm: IPv4 Address shows 10.10.10.10, Gateway 10.10.10.1
58. Test gateway: ping 10.10.10.1 — should get 4 replies

5.3 Install Active Directory Domain Services (AD DS)

[LEARN] LEARN WHY / APPRENDRE POURQUOI

WHAT IS ACTIVE DIRECTORY? AD is Microsoft's identity management system.
It stores all users (dr.martin, nurse.sophie), groups (Doctors, Nurses), and computers in a central database.
When you join a computer to the domain, AD controls who can log in and what they can access.
QU'EST-CE QU'ACTIVE DIRECTORY? C'est la base de données centrale des utilisateurs, groupes et ordinateurs.

59. Open Server Manager (it opens automatically on login, or Start → Server Manager)
60. Click 'Add roles and features' → Next → Next (Role-based or feature-based installation)
61. Select the local server DC01 → Next
62. In the Roles list, check: Active Directory Domain Services
63. A popup appears: click 'Add Features' → Next → Next → Next → Install
64. Wait for installation (5 minutes) — DO NOT close the wizard
65. When complete, click: 'Promote this server to a domain controller' (the yellow flag notification)

5.4 Promote to Domain Controller — Create hospital.fr Forest

[LEARN] LEARN WHY / APPRENDRE POURQUOI

WHAT IS A FOREST? In Active Directory, a 'forest' is the top-level container of everything.
Our forest is called hospital.fr — all users, computers, and policies live inside it.
We are creating a brand new forest (no existing AD) — called 'Add a new forest'.
QU'EST-CE QU'UNE FORÊT AD? La forêt hospital.fr est le conteneur de niveau supérieur de toute l'infrastructure AD.

66. Deployment Operation: select 'Add a new forest'
67. Root domain name: hospital.fr → click Next
68. Forest functional level: Windows Server 2016 (or 2022 if available) → Next
69. Domain Controller options: leave DNS server and Global Catalog checked
70. Set DSRM password: P@ssw0rd (this is a recovery password — write it down)

71. DNS Delegation: ignore the warning → Next
72. NetBIOS domain name: HOSPITAL (auto-filled) → Next → Next → Next → Install
73. Server will automatically reboot when complete
74. After reboot, log in as: HOSPITAL\Administrator / P@ssw0rd

✅ SUCCESS CHECK / VÉRIFICATION

CHECK: Server Manager shows AD DS and DNS in the left panel.
CHECK: Open DNS Manager (Tools → DNS) — you should see hospital.fr in Forward Lookup Zones.
CHECK: ipconfig /all should show 'Primary Dns Suffix: hospital.fr'

5.5 Configure DNS — Add Records for All VMs

Open DNS Manager: Server Manager → Tools → DNS → expand DC01 → Forward Lookup Zones → hospital.fr

Right-click hospital.fr → New Host (A or AAAA) → add each record:

Name (Host)	IP Address	Full DNS Name (auto-generated)
dc01	10.10.10.10	dc01.hospital.fr
app01	10.10.20.20	app01.hospital.fr
med01	10.10.20.21	med01.hospital.fr
fw01	10.10.10.1	fw01.hospital.fr
client01	10.10.30.30	client01.hospital.fr

Also add DNS Forwarders so DC01 can resolve internet names:

75. DNS Manager → right-click DC01 → Properties → Forwarders tab → Edit
76. Add: 8.8.8.8 (Google) and 1.1.1.1 (Cloudflare) → OK

5.6 Create Active Directory Users & Groups

Open Active Directory Users and Computers: Server Manager → Tools → Active Directory Users and Computers

Create Organizational Units (OUs)

77. Right-click hospital.fr → New → Organizational Unit → Name: Groups → OK
78. Repeat for: Users, Computers, Servers

Create Security Groups

Right-click the Groups OU → New → Group:

Group Name	Group Scope	Group Type	Purpose
G_Hospital_ITAdmins	Global	Security	IT administrators — full network access

Group Name	Group Scope	Group Type	Purpose
G_Hospital_Doctors	Global	Security	Medical staff — clinical VLAN access
G_Hospital_Nurses	Global	Security	Nursing staff — limited clinical access
G_Hospital_Staff	Global	Security	General staff — administrative VLAN

Create User Accounts

Right-click the Users OU → New → User:

First Name	Last Name	User Logon Name	Password	Member of Group	Role
Dr. Jean	Martin	dr.martin	Doctor@2024!	G_Hospital_Doctors	Chief of Medicine
Sophie	Bernard	nurse.sophie	Nurse@2024!	G_Hospital_Nurses	Head Nurse
Tech	Admin	admin.tech	Admin@2024!	G_Hospital_ITAdmins	IT Administrator
Marie	Reception	m.reception	Staff@2024!	G_Hospital_Staff	Receptionist

For each user, uncheck 'User must change password at next logon' for the lab environment.

5.7 Create Shared Folders for Clinical Staff

79. Create folder: C:\Shares\Clinical on DC01
80. Right-click Clinical folder → Properties → Sharing tab → Advanced Sharing
81. Check 'Share this folder', Share name: ClinicalShare → Permissions → Add G_Hospital_Doctors and G_Hospital_Nurses with Read/Change
82. Security tab → Edit → Add the same groups with appropriate NTFS permissions

✓ SUCCESS CHECK / VÉRIFICATION

VERIFY: Open CMD on DC01 → nslookup app01.hospital.fr → should return 10.10.20.20
VERIFY: nslookup google.com → should resolve (tests DNS forwarders to internet)
VERIFY: net user /domain → should list dr.martin, nurse.sophie, admin.tech

PHASE 4 — Install & Configure APP01

Ubuntu Web Application Server

6. APP01 — Ubuntu Server 22.04 (10.10.20.20)

ℹ INFO / NOTE

WHY: APP01 hosts the hospital web portal, database, and monitoring system.
GOAL: Install Ubuntu Server, configure static IP on VLAN 20, deploy LAMP stack.
OUTCOME: http://10.10.20.20 serves the Hospital.fr web portal. MySQL holds patient data.

6.1 Install Ubuntu Server 22.04

[LEARN] LEARN WHY / APPRENDRE POURQUOI

WHAT IS UBUNTU SERVER? Ubuntu Server is a free, enterprise Linux operating system from Canonical.
Unlike Ubuntu Desktop, it has no graphical interface — everything is done via command line.
This is how real production servers run: no GUI = less attack surface, less RAM usage, more stability.
UBUNTU SERVER est un Linux d'entreprise gratuit. Pas d'interface graphique = moins de RAM, plus de sécurité.

83. Attach Ubuntu Server 22.04 ISO to APP01 → Start APP01
84. Select: Try or Install Ubuntu Server → press Enter
85. Language: English (or French) → Done
86. Keyboard: select your layout → Done
87. Type of install: Ubuntu Server (not minimized) → Done
88. Network: skip for now (we will set static IP via netplan after install) → Done
89. Proxy: leave blank → Done
90. Mirror: leave default Ubuntu archive → Done
91. Storage: Use an entire disk → select the virtual disk → Done → Confirm destructive action: Continue
92. Profile setup:
    - Your name: Hospital Admin
    - Server name: app01
    - Username: hospitaladmin (or your preferred username)
    - Password: choose a strong password — note it carefully
93. OpenSSH: select 'Install OpenSSH server' ← IMPORTANT — check this box → Done
94. Snaps: skip (press Done without selecting anything)
95. Installation runs (5-15 minutes) → when complete: Reboot Now
96. Remove ISO when prompted → press Enter

6.2 Configure Static IP via Netplan

[LEARN] LEARN WHY / APPRENDRE POURQUOI

WHAT IS NETPLAN? Ubuntu uses Netplan to manage network configuration. Settings are written in YAML files.

YAML is space-sensitive — every indent MUST use exactly 2 spaces. Tabs will break the file.

We set a static IP so APP01 always has the same address. Other systems (DC01, pfSense) will refer to this IP in their DNS records.

QU'EST-CE QUE NETPLAN? Ubuntu utilise des fichiers YAML pour configurer le réseau. Les espaces sont critiques.

Log into APP01 console. First, find your interface name:

```
ip a
# Look for the interface name — usually 'enp0s3'. Note it exactly.
# You will see something like: '2: enp0s3: <BROADCAST,MULTICAST,UP...'
```

Edit the netplan configuration:

```
sudo nano /etc/netplan/00-installer-config.yaml
```

Delete all existing content and paste EXACTLY (replace enp0s3 with your actual interface name if different):

```
network:
  version: 2
  ethernets:
    enp0s3:
      dhcp4: no
      addresses:
        - 10.10.20.20/24
      routes:
        - to: default
          via: 10.10.20.1
      nameservers:
        addresses:
          - 10.10.10.10
          - 8.8.8.8
```

Save the file: Ctrl+X → Y → Enter. Then apply:

```
sudo chmod 600 /etc/netplan/00-installer-config.yaml
sudo netplan apply
```

Verify:

```
ip a show enp0s3
# Should show: inet 10.10.20.20/24
ip route show
# Should show: default via 10.10.20.1
ping 10.10.20.1 -c 4
# Should get 4 replies from pfSense gateway
ping 10.10.10.10 -c 4
# Should get 4 replies from DC01 — confirms inter-VLAN routing works
```

✅ SUCCESS CHECK / VÉRIFICATION

If ping 10.10.10.10 replies: VLAN inter-routing is working. This is the key test.

If ping fails: check pfSense firewall rules (Section 4.9) and promiscuous mode on APP01.

Si ping 10.10.10.10 répond: le routage inter-VLAN fonctionne — c'est le test clé.

6.3 Enable SSH — Copy-Paste Solution for Windows

[LEARN] LEARN WHY / APPRENDRE POURQUOI

PROBLEM: Ubuntu Server has no GUI. You cannot copy-paste into the VirtualBox console window easily.

SOLUTION: Use SSH from your Windows PC to connect to APP01. Then you can paste any command directly.

SSH (Secure Shell) is the standard way real sysadmins manage Linux servers remotely.

PROBLÈME: Ubuntu Server n'a pas de GUI. SOLUTION: Utilisez SSH depuis Windows — vous pouvez coller n'importe quelle commande.

Temporary Bridge for SSH Access (One-time setup)

97. Shut down APP01 completely
98. APP01 → Settings → Network → Adapter 1 → Attached to: Bridged Adapter → select your Wi-Fi or Ethernet in the dropdown
99. Start APP01 — it gets a home network IP (e.g. 192.168.1.x)
100. In APP01 console: ip a — note the 192.168.x.x address shown
101. On your Windows PC: press Win+X → Windows Terminal (or PowerShell)

```
ssh hospitaladmin@192.168.1.xxx
# Replace xxx with the actual IP from step 4. Type 'yes' on first connection.
```

102. You are now SSH'd in. Paste commands directly — right-click in PowerShell, or Ctrl+Shift+V in Windows Terminal
103. When done with all configuration: shut APP01 → return Adapter 1 to Internal Network: LabNet → Start

[TIP] TIP / CONSEIL

Use the Heredoc method to paste multi-line config files safely (prevents indent errors):

cat > /etc/netplan/00-installer-config.yaml << 'EOF'

[paste your config here]

EOF

Everything between the two EOF markers is written to the file exactly as pasted.

6.4 Install LAMP Stack

[LEARN] LEARN WHY / APPRENDRE POURQUOI

WHAT IS LAMP? LAMP = Linux + Apache + MySQL + PHP. This is the most common web server stack in the world.

Apache serves web pages. MySQL stores data (patient records, appointments). PHP generates dynamic HTML.

LAMP est la combinaison la plus utilisée au monde: Apache (serveur web) + MySQL (base de données) + PHP (langage).

```
sudo apt update && sudo apt upgrade -y
# Update package lists and upgrade all installed packages

sudo apt install -y apache2 mysql-server php libapache2-mod-php php-mysql php-cli php-gd php-curl php-mbstring php-xml
```

```
# Install Apache web server, MySQL database, and PHP with common modules

sudo systemctl enable apache2 mysql
# Enable services to start automatically on boot
sudo systemctl start apache2 mysql
```

Secure MySQL

```
sudo mysql_secure_installation
# Follow prompts: set root password, remove anonymous users, disallow remote root, remove
test DB
# Answer 'Y' to all questions for a secure installation
```

6.5 Create Hospital Database

```
sudo mysql -u root -p
# Enter the MySQL root password you set in 6.4

CREATE DATABASE hospital_portal CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
CREATE USER 'hospital_app'@'localhost' IDENTIFIED BY 'H0sp1tal@2024';
GRANT ALL PRIVILEGES ON hospital_portal.* TO 'hospital_app'@'localhost';
FLUSH PRIVILEGES;

USE hospital_portal;

CREATE TABLE patients (
  id INT AUTO_INCREMENT PRIMARY KEY,
  patient_id VARCHAR(20) UNIQUE NOT NULL,
  first_name VARCHAR(50) NOT NULL,
  last_name VARCHAR(50) NOT NULL,
  date_of_birth DATE,
  assigned_doctor VARCHAR(100),
  admission_date DATE,
  status ENUM('Admitted','Discharged','Outpatient') DEFAULT 'Outpatient'
);

CREATE TABLE appointments (
  id INT AUTO_INCREMENT PRIMARY KEY,
  patient_id VARCHAR(20),
  doctor_name VARCHAR(100),
  appointment_date DATETIME,
  department VARCHAR(50),
  notes TEXT
);

# Insert sample data for the demo
INSERT INTO patients VALUES
  (NULL,'P001','Marie','Dupont','1985-03-15','dr.martin','2024-01-10','Admitted'),
  (NULL,'P002','Jean','Bernard','1972-07-22','dr.martin','2024-01-12','Outpatient'),
  (NULL,'P003','Sophie','Martin','1990-11-05','nurse.sophie','2024-01-
14','Outpatient');

EXIT;
```

6.6 Deploy Hospital Web Portal

```
sudo mkdir -p /var/www/html/hospital
sudo nano /var/www/html/hospital/index.php
```

Paste the following PHP application:

```
<?php
session_start();
$host='localhost'; $db='hospital_portal';
$user='hospital_app'; $pass='H0sp1tal@2024';
$conn=new mysqli($host,$user,$pass,$db);

$patients=[]; $result=$conn->query('SELECT * FROM patients');
while($row=$result->fetch_assoc()) $patients[]=$row;
?><!DOCTYPE html><html><head>
<title>Hospital.fr Portal</title>
<style>
body{font-family:Arial,sans-serif;background:#f0f8ff;margin:0}
.header{background:#003366;color:#fff;padding:20px 40px}
.header h1{margin:0;font-size:28px}
.header p{margin:4px 0;opacity:.8;font-size:14px}
.nav{background:#0055aa;padding:10px 40px}
.nav a{color:#fff;margin-right:20px;text-decoration:none;font-size:14px}
.container{padding:30px 40px}
.card{background:#fff;border-radius:8px;padding:24px;margin-bottom:20px;box-shadow:0
2px 8px rgba(0,0,0,.1)}
.badge{padding:4px 12px;border-radius:12px;font-size:12px;font-weight:bold}
.admitted{background:#ffe0e0;color:#c00}
.outpatient{background:#e0f0e0;color:#060}
.discharged{background:#e0e0e0;color:#444}
table{width:100%;border-collapse:collapse}
th{background:#003366;color:#fff;padding:10px;text-align:left}
td{padding:10px;border-bottom:1px solid #eee}
tr:hover td{background:#f5f9ff}
</style></head><body>
<div class='header'>
  <h1>[HOSPITAL] Hospital.fr — Healthcare Management Portal</h1>
  <p>Server: <?=gethostname()?>  |  VLAN 20 — Clinical Network  | 
<?=date('Y-m-d H:i')?></p>
</div>
<div class='nav'>
  <a href='/hospital/'>Dashboard</a>
  <a href='/hospital/?tab=patients'>Patients</a>
  <a href='http://<?=$_SERVER["HTTP_HOST"]?>:8042' target='_blank'>PACS Imaging</a>
  <a href='/zabbix' target='_blank'>Monitoring</a>
</div>
<div class='container'>
  <div class='card'><h2>System
Status</h2><table><tr><th>Service</th><th>IP:Port</th><th>Status</th></tr>
  <?php foreach(['Domain Controller'=>'10.10.10.10:389','Web
Server'=>'10.10.20.20:80','PACS'=>'10.10.20.21:8042'] as $n=>$h){
    list($ip,$port)=explode(':',$h);
    $ok=@fsockopen($ip,(int)$port,$e,$s,1);
    echo "<tr><td>$n</td><td>$h</td><td>".($ok?"<span class='badge
admitted'>ONLINE</span>":"<span class='badge
discharged'>OFFLINE</span>")."</td></tr>";
    if($ok)fclose($ok); } ?></table></div>
  <div class='card'><h2>Patient List (<?=count($patients)?> records)</h2><table>
```

```
<tr><th>ID</th><th>Name</th><th>DOB</th><th>Doctor</th><th>Admitted</th><th>Status</th
></tr>
  <?php foreach($patients as $p){ $cls=strtolower($p['status']);
    echo "<tr><td>{$p['patient_id']}</td><td>{$p['first_name']}
{$p['last_name']}</td>";
    echo "<td>{$p['date_of_birth']}</td><td>{$p['assigned_doctor']}</td>";
    echo "<td>{$p['admission_date']}</td><td><span class='badge
$cls'>{$p['status']}</span></td></tr>"; } ?>
  </table></div>
</div></body></html>
```

```
sudo chown -R www-data:www-data /var/www/html/hospital
sudo systemctl restart apache2
```

✅ SUCCESS CHECK / VÉRIFICATION

VERIFY: From CLIENT01 browser, navigate to http://10.10.20.20/hospital/
You should see the Hospital.fr portal with patient list and system status indicators.
VÉRIFICATION: Depuis le navigateur de CLIENT01, naviguez vers http://10.10.20.20/hospital/

PHASE 5 — Install & Configure MED01

Systèmes Médicaux & PACS

7. MED01 — Medical Systems Server (10.10.20.21)

INFO / NOTE

WHY: MED01 simulates the medical imaging (PACS) server found in every real hospital.
GOAL: Install Ubuntu Server, configure static IP on VLAN 20, deploy Orthanc PACS.
OUTCOME: http://10.10.20.21:8042 serves the Orthanc DICOM image viewer.

7.1 Install Ubuntu Server 22.04 on MED01

Repeat the same installation process as APP01 (Section 6.1), with these differences:

Setting	APP01 value	MED01 value
VM Name in VirtualBox	APP01	MED01
Server name during install	app01	med01
Final static IP	10.10.20.20	10.10.20.21
Gateway	10.10.20.1	10.10.20.1 (same VLAN)
DNS	10.10.10.10	10.10.10.10 (same DC01)

7.2 Configure Static IP on MED01

```
sudo nano /etc/netplan/00-installer-config.yaml
```

```
network:
  version: 2
  ethernets:
    enp0s3:
      dhcp4: no
      addresses:
        - 10.10.20.21/24
      routes:
        - to: default
          via: 10.10.20.1
      nameservers:
        addresses:
          - 10.10.10.10
          - 8.8.8.8
```

```
sudo chmod 600 /etc/netplan/00-installer-config.yaml
sudo netplan apply
```

Verify connectivity:

```
ping 10.10.20.1 -c 3
# Gateway — should reply
ping 10.10.10.10 -c 3
# DC01 — confirms inter-VLAN routing
ping 10.10.20.20 -c 3
# APP01 — confirms same-VLAN communication
```

7.3 Install Orthanc PACS

[LEARN] LEARN WHY / APPRENDRE POURQUOI

WHAT IS PACS? Picture Archiving and Communication System — the database and viewer for medical images. Every X-ray, MRI, CT scan in a hospital is stored in DICOM format and managed by a PACS server.
Orthanc is a free, open-source PACS server. It is used in real hospitals worldwide.
QU'EST-CE QU'UN PACS? Système d'archivage et de communication d'images médicales. Orthanc est un serveur PACS open-source utilisé dans de vrais hôpitaux.

```
sudo apt update
sudo apt install -y software-properties-common wget curl gnupg

# Add Orthanc official repository
wget -qO - https://debian.orthanc-labs.com/archive.key | sudo apt-key add -
echo 'deb https://debian.orthanc-labs.com/ jammy main' | sudo tee
/etc/apt/sources.list.d/orthanc.list

sudo apt update
sudo apt install -y orthanc orthanc-webviewer

sudo systemctl enable orthanc
sudo systemctl start orthanc
```

Verify Orthanc is running:

```
sudo systemctl status orthanc
# Should show: Active: active (running)

curl http://localhost:8042/system
# Should return JSON: {"ApiVersion":..., "OrthancApiVersion":..., "Version":...}
```

✅ SUCCESS CHECK / VÉRIFICATION

VERIFY: From CLIENT01 browser → http://10.10.20.21:8042
You should see the Orthanc Explorer web interface with login fields.
Default credentials: orthanc / orthanc
Upload a free DICOM test image: visit dicom.org/dicom-standard/sample-images to download samples.
VÉRIFICATION: Depuis CLIENT01 → http://10.10.20.21:8042 — vous devriez voir l'interface Orthanc.

7.4 Install Zabbix Monitoring Agent on MED01

ℹ INFO / NOTE

The Zabbix agent on MED01 reports health metrics back to the Zabbix server on APP01.

```
wget https://repo.zabbix.com/zabbix/6.0/ubuntu/pool/main/z/zabbix-release/zabbix-
release_6.0-4+ubuntu22.04_all.deb
sudo dpkg -i zabbix-release_6.0-4+ubuntu22.04_all.deb
sudo apt update
sudo apt install -y zabbix-agent

# Configure agent to report to APP01 (Zabbix server)
sudo nano /etc/zabbix/zabbix_agentd.conf
# Find and change these lines:
#   Server=10.10.20.20
#   ServerActive=10.10.20.20
#   Hostname=MED01

sudo systemctl enable zabbix-agent && sudo systemctl start zabbix-agent
```

PHASE 6 — Install Client & Guest VMs

Postes de Travail et Accès Invités

8. CLIENT01 — Staff Workstation & GUEST01

8.1 Install Windows 11 on CLIENT01

[LEARN] LEARN WHY / APPRENDRE POURQUOI

CLIENT01 represents any doctor or nurse workstation in the hospital. By joining it to the hospital.fr domain, users can log in with their AD credentials from any domain computer.

CLIENT01 représente le poste de travail d'un médecin ou d'une infirmière. En le joignant au domaine, les utilisateurs peuvent se connecter avec leurs identifiants AD.

104. Attach Windows 11 ISO to CLIENT01 → Start VM
105. Install Windows 11 normally — choose Windows 11 Pro (required for domain join)
106. During setup when asked for a Microsoft account: click 'I don't have internet' → 'Continue with limited setup'
107. Create a local account: localadmin / LocalPass@2024
108. Complete installation and reach the desktop

8.2 Set Static IP on CLIENT01

Settings → Network & Internet → Ethernet → IP assignment → Edit → Manual:

Field	Value
IPv4	On
IP Address	10.10.30.30
Subnet Prefix Length	24
Gateway	10.10.30.1
Preferred DNS	10.10.10.10
Alternate DNS	8.8.8.8

Verify: open CMD → ipconfig /all → confirm 10.10.30.30. Then:

```
ping 10.10.30.1
# VLAN 30 gateway — should reply
ping 10.10.10.10
# DC01 — MUST reply before attempting domain join
ping 10.10.20.20
# APP01 — tests cross-VLAN firewall rules
```

8.3 Join CLIENT01 to hospital.fr Domain

[LEARN] LEARN WHY / APPRENDRE POURQUOI

WHAT IS A DOMAIN JOIN? When a computer joins the domain, it trusts the Domain Controller for authentication. After joining, any user in Active Directory can log into that computer using their domain credentials.
The computer also receives Group Policy from DC01 — automatically enforcing security settings.
LA JONCTION AU DOMAINE permet à tout utilisateur AD de se connecter sur ce poste et reçoit automatiquement les politiques de groupe.

109. Settings → System → About → scroll to 'Domain or workgroup' → click Change
110. Or: press Win+R → type: sysdm.cpl → Computer Name tab → click Change
111. Under 'Member of', select Domain → type: hospital.fr → click OK
112. Authentication prompt appears — enter: HOSPITAL\Administrator / P@ssw0rd
113. 'Welcome to the hospital.fr domain!' message → OK → OK → Restart Now
114. After reboot, at login screen: click Other User → enter: HOSPITAL\dr.martin / Doctor@2024!

✓ SUCCESS CHECK / VÉRIFICATION

VERIFY: Log in as HOSPITAL\dr.martin — desktop loads with domain profile.
VERIFY: Open CMD → type: whoami — should show: hospital\dr.martin
VERIFY: Open CMD → type: nslookup app01.hospital.fr — should return 10.10.20.20
VERIFY: Open browser → http://10.10.20.20/hospital/ — Hospital Portal should load
VÉRIFICATION: Connectez-vous en tant que HOSPITAL\dr.martin — le bureau du domaine se charge.

8.4 Install GUEST01 (IoT / Guest VM)

Install Windows 11 on GUEST01 following the same process as CLIENT01, with these differences:

Setting	CLIENT01	GUEST01
Purpose	Domain-joined staff workstation	Isolated guest/IoT simulation
IP Address	10.10.30.30	10.10.40.40
VLAN	VLAN 30 — Staff	VLAN 40 — IoT/Guest
Domain Join	Yes — hospital.fr	NO — leave as workgroup
DNS	10.10.10.10 (DC01)	8.8.8.8 (internet DNS only)
Expected access	Clinical apps, domain shares	Internet only — no internal access

✓ SUCCESS CHECK / VÉRIFICATION

VERIFY from GUEST01: ping 8.8.8.8 — should succeed (internet works)
VERIFY from GUEST01: ping 10.10.20.20 — should FAIL (blocked by pfSense)
This proves VLAN isolation: guests cannot reach hospital systems.
VÉRIFICATION depuis GUEST01: ping 10.10.20.20 doit ÉCHOUER — prouve l'isolation VLAN.

PHASE 7 — Deploy Monitoring & Services

Déployer la Surveillance et les Services

9. Deploy Zabbix Monitoring on APP01

ℹ INFO / NOTE

WHY: A production network without monitoring is blind. You must know when servers are down or slow.
GOAL: Install Zabbix on APP01 — it monitors all VMs and sends alerts on problems.
OUTCOME: Real-time dashboard at http://10.10.20.20/zabbix showing the health of the entire hospital network.

9.1 Install Zabbix Server on APP01

[LEARN] LEARN WHY / APPRENDRE POURQUOI

WHAT IS ZABBIX? Zabbix is a free, enterprise-grade monitoring platform used by thousands of companies.
It collects metrics (CPU, RAM, disk, network) from agents installed on each server.
When something breaks — a server goes offline, disk fills up — Zabbix sends an alert by email or SMS.
ZABBIX est une plateforme de surveillance gratuite. Elle collecte des métriques de tous les serveurs et envoie des alertes.

SSH into APP01 (10.10.20.20) and run:

```
wget https://repo.zabbix.com/zabbix/6.0/ubuntu/pool/main/z/zabbix-release/zabbix-release_6.0-4+ubuntu22.04_all.deb
sudo dpkg -i zabbix-release_6.0-4+ubuntu22.04_all.deb
sudo apt update
sudo apt install -y zabbix-server-mysql zabbix-frontend-php zabbix-apache-conf zabbix-sql-scripts zabbix-agent
```

9.2 Configure Zabbix Database

```
sudo mysql -u root -p
# Enter your MySQL root password

CREATE DATABASE zabbix CHARACTER SET utf8mb4 COLLATE utf8mb4_bin;
CREATE USER 'zabbix'@'localhost' IDENTIFIED BY 'Z@bb1x2024!';
GRANT ALL PRIVILEGES ON zabbix.* TO 'zabbix'@'localhost';
FLUSH PRIVILEGES;
EXIT;

# Import the Zabbix database schema
zcat /usr/share/zabbix-sql-scripts/mysql/server.sql.gz | mysql --default-character-set=utf8mb4 -uzabbix -p zabbix
# This takes 1-3 minutes — wait for it to complete

# Set the database password in Zabbix config
sudo sed -i 's/# DBPassword=/DBPassword=Z@bb1x2024!/' /etc/zabbix/zabbix_server.conf
```

```
sudo systemctl restart zabbix-server zabbix-agent apache2
sudo systemctl enable zabbix-server zabbix-agent apache2
```

9.3 Complete Zabbix Setup via Web Browser

115. From CLIENT01 browser: http://10.10.20.20/zabbix
116. Click Next until you reach the Database Configuration page
117. Enter: Host=localhost, Port=3306, Database=zabbix, User=zabbix, Password=Z@bb1x2024!
118. Click Next → configure Server details: Name=Hospital.fr Network → Next → Finish
119. Log in: Admin / zabbix
120. Change password: top-right user icon → Profile → change password immediately

Add Hosts to Monitoring

121. Configuration → Hosts → Create host
122. Add each VM: DC01 (10.10.10.10), MED01 (10.10.20.21), CLIENT01 (10.10.30.30)
123. For each host: Hostname = VM name, IP = VM IP, Port = 10050, link the 'Linux by Zabbix agent' template
124. Install zabbix-agent on DC01 and MED01 as shown in Section 7.4

✓ SUCCESS CHECK / VÉRIFICATION

VERIFY: Monitoring → Hosts — all hosts show Green availability icon.
VERIFY: Monitoring → Latest data — you should see CPU, memory, disk metrics streaming in.
VÉRIFICATION: Monitoring → Hôtes — tous les hôtes affichent une icône de disponibilité verte.

PHASE 8 — Security Configuration

Configuration de la Sécurité

10. Security Configuration / Configuration de la Sécurité

10.1 pfSense Production Firewall Rules

[LEARN] LEARN WHY / APPRENDRE POURQUOI

SECURITY MODEL: 'Least Privilege' — each VLAN only gets access to what it absolutely needs.
Think of it like hospital badge access: a nurse accesses patient floors, not the pharmacy safe.
We now REPLACE the temporary 'Allow All' rules with precise, purpose-built rules.
MODÈLE DE SÉCURITÉ: 'Moindre Privilège' — chaque VLAN n'accède qu'à ce dont il a besoin.
Nous remplaçons maintenant les règles temporaires 'Allow All' par des règles précises.

Step 1 — Delete Temporary Rules

Firewall → Rules → for each VLAN tab (MGMT, CLINICAL, ADMIN, GUEST): find the 'TEMP Allow All' rule → click the trash icon → Save → Apply Changes

Step 2 — MGMT_VLAN10 Rules (Management)

Protocol	Source	Destination	Ports	Action	Description
Any	MGMT_VLAN10 net	Any	Any	Pass	IT Admins: full access to everything

Step 3 — CLINICAL_VLAN20 Rules

Protocol	Source	Destination	Ports	Action	Description
TCP/UDP	CLINICAL_VLAN20 net	10.10.10.10	53,88,389,636	Pass	DNS, Kerberos, LDAP to DC01
TCP	CLINICAL_VLAN20 net	CLINICAL_VLAN20 net	Any	Pass	Clinical servers talk to each other
TCP	CLINICAL_VLAN20 net	WAN	80,443	Pass	Internet access for updates
Any	CLINICAL_VLAN20 net	Any	Any	Block	Default deny everything else

Step 4 — ADMIN_VLAN30 Rules (Staff)

Protocol	Source	Destination	Ports	Action	Description
TCP/UDP	ADMIN_VLAN30 net	10.10.10.10	53,88,389,445	Pass	Auth + file shares on DC01
TCP	ADMIN_VLAN30 net	10.10.20.0/24	80,443,8042	Pass	Web portal + PACS on VLAN 20
TCP	ADMIN_VLAN30 net	WAN	80,443	Pass	Internet access
Any	ADMIN_VLAN30 net	MGMT_VLAN10 net	Any	Block	Staff cannot access management VLAN
Any	ADMIN_VLAN30 net	Any	Any	Block	Default deny

Step 5 — GUEST_VLAN40 Rules

Protocol	Source	Destination	Ports	Action	Description
TCP	GUEST_VLAN40 net	WAN	80,443	Pass	Internet browsing only
Any	GUEST_VLAN40 net	!WAN	Any	Block	Block ALL internal access — critical

[CRITICAL] CRITICAL / CRITIQUE

The GUEST VLAN block rule destination uses '!WAN' (NOT WAN) meaning: block access to anything that is NOT the internet.

In pfSense: Destination → select 'Invert match' checkbox → select 'WAN net'. This blocks all internal traffic.

La règle de blocage GUEST utilise 'NOT WAN' pour bloquer tout accès non-internet.

10.2 Group Policy Objects (GPOs) on DC01

[LEARN] LEARN WHY / APPRENDRE POURQUOI

WHAT ARE GPOs? Group Policy Objects are configuration rules that Active Directory pushes to all domain computers automatically.

A new workstation joining the domain instantly gets all policies — no manual configuration needed.

This is how hospitals enforce HIPAA/HDS compliance across hundreds of computers at once.

LES GPO sont des règles de configuration que Active Directory pousse automatiquement à tous les ordinateurs du domaine.

On DC01: Server Manager → Tools → Group Policy Management → right-click hospital.fr → Create a GPO:

GPO 1 — Hospital Security Baseline

Setting Path	Policy Name	Value	Purpose
Computer Config → Policies → Windows Settings → Security Settings → Account Policies	Minimum password length	10 characters	Strong passwords for all accounts

Setting Path	Policy Name	Value	Purpose
Computer Config → ... → Account Lockout Policy	Lockout threshold	5 attempts	Brute-force protection
Computer Config → ... → Account Lockout Policy	Lockout duration	30 minutes	Auto-unlock after 30 min
User Config → Policies → Admin Templates → Control Panel → Personalization	Screen saver timeout	300 seconds	Auto-lock after 5 minutes idle
User Config → ... → Personalization	Password protect screen saver	Enabled	Must re-enter password to unlock

GPO 2 — Map Clinical Drives for Doctors

User Config → Preferences → Windows Settings → Drive Maps → New → Mapped Drive:

Field	Value
Action	Create
Location	\\\\dc01.hospital.fr\\ClinicalShare
Drive Letter	Z:
Label	Clinical Files
Item Level Targeting	Security Group: G_Hospital_Doctors

Link GPOs: right-click hospital.fr → Link an Existing GPO → select each GPO. Test: run gpupdate /force on CLIENT01

10.3 Active Directory Security Groups — RBAC

[LEARN] LEARN WHY / APPRENDRE POURQUOI

RBAC = Role-Based Access Control. Each user gets exactly the access their role requires — nothing more.
Dr. Martin is a Doctor: he gets access to PACS, patient records, clinical shares.
Reception staff gets appointment booking but NOT medical images.
RBAC = Contrôle d'Accès Basé sur les Rôles. Chaque utilisateur obtient exactement l'accès requis par son rôle.

User	AD Group	Access Level	VLAN	Can access
admin.tech	G_Hospital_ITAdmins	Full	VLAN 10	All systems, pfSense, servers
dr.martin	G_Hospital_Doctors	Clinical	VLAN 30→20	Portal, PACS, clinical shares, domain
nurse.sophie	G_Hospital_Nurses	Limited Clinical	VLAN 30→20	Portal read-only, PACS viewer
m.reception	G_Hospital_Staff	Administrative	VLAN 30	Appointments portal only
GUEST01	(no domain)	Internet Only	VLAN 40	8.8.8.8 — nothing internal

PHASE 9 — Testing, Validation & Demo

Tests, Validation et Démonstration

11. Testing, Validation & Demo Scenarios

ℹ INFO / NOTE

This section is your complete verification checklist AND your demo script for shareholders.
Run ALL tests before any presentation. If a test fails, use Section 12 troubleshooting to fix it.
Cette section est votre liste de vérification et votre script de démonstration pour les actionnaires.

11.1 Master Connectivity Test Matrix / Matrice de Tests

Test #	From VM	Command	Expected Result	Result
1	APP01	ping 10.10.20.1 -c 4	4 replies — VLAN 20 gateway	[ ] ✓
2	APP01	ping 10.10.10.10 -c 4	4 replies — DC01 inter-VLAN	[ ] ✓
3	MED01	ping 10.10.20.1 -c 4	4 replies — VLAN 20 gateway	[ ] ✓
4	MED01	ping 10.10.10.10 -c 4	4 replies — DC01 inter-VLAN	[ ] ✓
5	MED01	ping 10.10.20.20 -c 4	4 replies — APP01 same VLAN	[ ] ✓
6	DC01	ping 10.10.10.1 -n 4	4 replies — VLAN 10 gateway	[ ] ✓
7	DC01	ping 10.10.20.20 -n 4	4 replies — APP01 inter-VLAN	[ ] ✓
8	CLIENT01	ping 10.10.30.1 -n 4	4 replies — VLAN 30 gateway	[ ] ✓
9	CLIENT01	ping 10.10.10.10 -n 4	4 replies — DC01	[ ] ✓
10	CLIENT01	ping 10.10.20.20 -n 4	4 replies — APP01	[ ] ✓
11	APP01	nslookup dc01.hospital.fr 10.10.10.10	Returns 10.10.10.10	[ ] ✓
12	CLIENT01	nslookup app01.hospital.fr	Returns 10.10.20.20	[ ] ✓
13	APP01	ping 8.8.8.8 -c 4	4 replies — Internet works	[ ] ✓
14	GUEST01	ping 8.8.8.8 -n 4	4 replies — Internet works	[ ] ✓
15	GUEST01	ping 10.10.20.20 -n 4	Request timed out — BLOCKED	[ ] ✓

11.2 Service Availability Checks

Service	URL to test	Expected	Check
pfSense Admin	https://10.10.10.1 (from DC01 browser)	Login page appears	[ ]
Hospital Portal	http://10.10.20.20/hospital/ (from CLIENT01)	Portal with patient list loads	[ ]
Orthanc PACS	http://10.10.20.21:8042 (from CLIENT01)	Orthanc Explorer page	[ ]
Zabbix Monitor	http://10.10.20.20/zabbix (from CLIENT01)	Login page, then dashboard	[ ]
Domain Auth	Log into CLIENT01 as HOSPITAL\dr.martin	Successful domain login	[ ]
DNS Resolution	nslookup hospital.fr on CLIENT01	Resolves to 10.10.10.10	[ ]

11.3 Shareholder Demo Script — 15 Minutes

[LEARN] LEARN WHY / APPRENDRE POURQUOI

This is your presentation script. Practice it twice before the real demo.
Each scenario takes about 3 minutes. The entire demo is 15 minutes.
C'est votre script de présentation. Entrainez-vous deux fois avant la vraie démonstration.

[>>] Scenario 1 (3 min): Network Architecture Overview

Show: pfSense web UI (opened on DC01 browser, https://10.10.10.1)

- Status → Dashboard: show all 4 VLAN interfaces (MGMT, CLINICAL, ADMIN, GUEST) with green UP status
- Interfaces → Assignments: show the 4 VLANs on em1 with their 10.10.x.1 IPs
- Status → DHCP Leases: show each VM with its correct IP on the correct VLAN

Say: 'This is our segmented hospital network. Each department lives on its own VLAN, completely isolated. A ransomware attack on the guest network cannot reach patient data — they are physically separated at the network level.'

[>>] Scenario 2 (3 min): Clinician Login & Data Access

Show: CLIENT01 (VLAN 30)

- Log in as HOSPITAL\dr.martin — wait for domain desktop
- Run: whoami → shows hospital\dr.martin
- Open browser → http://10.10.20.20/hospital/ → Hospital Portal loads with patient list
- Click PACS link → http://10.10.20.21:8042 → Orthanc opens
- Show drive Z: automatically mapped (ClinicalShare from DC01)

Say: 'Dr. Martin logs in from any workstation in the hospital and automatically gets access to his patients, imaging system, and shared files — all secured by Active Directory through VLAN 10, even though he is sitting on VLAN 30.'

[>>] Scenario 3 (3 min): Medical Imaging (PACS Demo)

Show: Orthanc at http://10.10.20.21:8042 from CLIENT01

- Log in: orthanc / orthanc
- Upload a DICOM test image (pre-downloaded from dicom.org)
- Click the uploaded study → open in web viewer → image displays in browser

Say: 'MED01 runs Orthanc, an open-source PACS system used in real hospitals. Every medical image — X-ray, MRI, CT scan — is stored in DICOM format. Doctors access images from their workstation browser without installing any software.'

[>>] Scenario 4 (3 min): Security Incident Demonstration

Show: GUEST01 (VLAN 40) — simulating a patient or visitor device

- Open CMD on GUEST01 → ping 10.10.20.21 → Request timed out (blocked)
- ping 10.10.10.10 → Request timed out (blocked)
- ping 8.8.8.8 → Replies (internet works)
- On pfSense: Status → System Logs → Firewall → show the blocked packets in real time

Say: 'A visitor connecting to our guest WiFi gets internet access but is completely blocked from all hospital systems. Every blocked attempt is logged. This satisfies GDPR, HIPAA, and HDS compliance requirements for patient data isolation.'

[>>] Scenario 5 (3 min): IT Admin & Monitoring

- Show Zabbix dashboard: http://10.10.20.20/zabbix — all hosts green
- Monitoring → Latest Data → show CPU, memory, disk metrics for all VMs
- Show pfSense firewall logs with blocked connections
- Show Zabbix trigger configuration for automated alerts

Say: 'The IT team has full visibility across all systems from the management VLAN. Zabbix monitors every server 24/7 and sends alerts before problems escalate. This is the standard in enterprise hospital IT management.'

11.4 Quick Pre-Presentation Boot Sequence

Run this sequence 10 minutes before the demo:

Step	Action	Wait time	Verify
1	Start FW01 (pfSense)	60 seconds	Console shows LAN: 10.10.10.1
2	Start DC01 (Windows Server)	3 minutes	Ctrl+Alt+Del works, CMD responds
3	Start APP01 (Ubuntu)	30 seconds	Login prompt appears
4	Start MED01 (Ubuntu)	30 seconds	Login prompt appears
5	Start CLIENT01 (Windows 11)	2 minutes	Login screen appears
6	Open browser on DC01	—	Navigate to https://10.10.10.1 — pfSense loads
7	Verify all services	—	Run Test #1 through #6 from Section 11.2

PHASE 10 — Backup & Recovery

Sauvegarde et Récupération

12. Backup & Recovery / Sauvegarde et Récupération

12.1 VirtualBox Snapshots

[LEARN] LEARN WHY / APPRENDRE POURQUOI

WHAT IS A SNAPSHOT? A snapshot saves the exact state of a VM at a specific moment in time.
If something breaks after a change, you restore the snapshot and you are instantly back to the working state.
Take snapshots BEFORE every major phase. Storage is cheap — regret is expensive.
QU'EST-CE QU'UN SNAPSHOT? Il sauvegarde l'état exact d'une VM. Si quelque chose casse, vous restaurez et revenez à l'état de fonctionnement.

When to take snapshot	Suggested name	What it protects
Before Phase 1 (VMs created)	00-VMs-Created-Empty	Starting point — all 6 VMs configured
After pfSense VLAN setup	01-pfSense-VLANs-Complete	Working multi-VLAN router
After DC01 domain promotion	02-DC01-Domain-Live	Active Directory, DNS working
After APP01 + MED01 configured	03-Servers-Deployed	LAMP + Orthanc running
After CLIENT01 domain join	04-Client-Domain-Joined	Full working lab
After security rules applied	05-Security-Hardened	Production security rules active
Final working state	FINAL-Production-Baseline	The presentation-ready state

How to take snapshot: Right-click any VM in VirtualBox Manager → Snapshots → Take → enter the name above → OK

12.2 Automated Backup Script for APP01

SSH into APP01 and create this backup script:

```
sudo nano /usr/local/bin/backup_hospital.sh
```

```
#!/bin/bash
# Hospital.fr Automated Backup Script
# Runs daily at 2:00 AM via cron

BACKUP_DIR="/var/backups/hospital"
DATE=$(date +%Y%m%d_%H%M%S)
mkdir -p $BACKUP_DIR

# Backup web application files
tar -czf $BACKUP_DIR/webapp_$DATE.tar.gz /var/www/html/ 2>/dev/null
```

```
# Backup hospital database
mysqldump -u hospital_app -p'H0sp1tal@2024' hospital_portal > $BACKUP_DIR/db_$DATE.sql
2>/dev/null

# Backup Zabbix database (smaller — export config only)
mysqldump -u zabbix -p'Z@bb1x2024!' zabbix > $BACKUP_DIR/zabbix_$DATE.sql 2>/dev/null

# Backup network configuration
cp /etc/netplan/00-installer-config.yaml $BACKUP_DIR/netplan_$DATE.yaml

# Keep only last 7 days of backups
find $BACKUP_DIR -type f -mtime +7 -delete

echo "[$(date '+%Y-%m-%d %H:%M:%S')] Backup completed successfully" >>
/var/log/hospital_backup.log
```

```
sudo chmod +x /usr/local/bin/backup_hospital.sh
# Schedule daily at 2:00 AM
(crontab -l 2>/dev/null; echo '0 2 * * * /usr/local/bin/backup_hospital.sh') | crontab
-
# Verify it is scheduled
crontab -l
```

12.3 Restore Procedures

Restore database from backup

```
mysql -u hospital_app -p'H0sp1tal@2024' hospital_portal <
/var/backups/hospital/db_20240115_020001.sql
```

Restore web files from backup

```
sudo tar -xzf /var/backups/hospital/webapp_20240115_020001.tar.gz -C /
sudo systemctl restart apache2
```

Full VM restore from VirtualBox snapshot

125. Select the VM in VirtualBox Manager → click Snapshots tab
126. Right-click the snapshot to restore → Restore Snapshot → check 'Create a snapshot of the current state' if needed
127. Click Restore → Start the VM

✅ SUCCESS CHECK / VÉRIFICATION

Test your backup: run the script manually first: sudo /usr/local/bin/backup_hospital.sh
Check: ls /var/backups/hospital/ — you should see .tar.gz and .sql files
Check: tail /var/log/hospital_backup.log — should show 'Backup completed successfully'

PHASE 11 — Future Advancements

Évolutions et Extensions Futures

13. Future Advancements / Évolutions Futures

[EN] ENGLISH

This section is your technical growth roadmap. Each advancement is listed with its difficulty level, estimated time, and professional impact. They are ordered by recommended implementation priority.

[FR] FRANÇAIS

Cette section est votre feuille de route de croissance technique. Chaque évolution est listée avec son niveau de difficulté, le temps estimé et l'impact professionnel.

13.1 Priority Roadmap / Feuille de Route Prioritaire

#	Advancement	Difficulty	Time	Demo Value	Course Value
1	Add DC02 (High-Availability Domain)	Easy	1-2 hrs	★★★★	HA concepts
2	OpenEMR (Full Electronic Medical Record)	Medium	3-4 hrs	★★★★★	Clinical apps
3	Add 3 more workstations (DOCTOR02, NURSE01, LAB01)	Easy	30 min each	★★★	Scalability
4	Group Policy — Software Deployment	Medium	2 hrs	★★★	Enterprise AD
5	Suricata IDS/IPS on pfSense	Medium	2-3 hrs	★★★★	Security depth
6	OpenVPN Remote Doctor Access	Medium	2-3 hrs	★★★★	Remote access
7	Grafana Dashboards (visual monitoring)	Easy	1-2 hrs	★★★	Visualization
8	Mirth Connect HL7 Engine	Advanced	4-5 hrs	★★★	Healthcare interop
9	pfSense CARP HA Failover	Advanced	3-4 hrs	★★	HA networking
10	Nextcloud File Sharing Platform	Medium	2 hrs	★★★	Collaboration

13.2 Add DC02 — Secondary Domain Controller

[LEARN] LEARN WHY / APPRENDRE POURQUOI

WHY: If DC01 fails, authentication stops for every user in the hospital. DC02 is the redundant backup.
When both DCs are running, AD replicates automatically — same users, same passwords, same policies on both.
In real hospitals, multiple DCs per site are mandatory for business continuity.
POURQUOI: Si DC01 tombe en panne, l'authentification s'arrête. DC02 est la sauvegarde redondante.

128. Create new VM: Windows Server 2022, 2 GB RAM, VLAN 10, static IP 10.10.10.11
129. Install Windows Server 2022 → join domain hospital.fr with HOSPITAL\Administrator
130. Server Manager → Add Roles → Active Directory Domain Services → Install
131. Click yellow flag → Promote this server to DC → 'Add a domain controller to an existing domain'
132. Domain: hospital.fr → credentials: HOSPITAL\Administrator
133. Leave all options as-is → Install → reboot
134. After reboot: test by powering off DC01 → verify domain login still works through DC02

✅ SUCCESS CHECK / VÉRIFICATION

DC01 OFF + login as HOSPITAL\dr.martin on CLIENT01 = SUCCESS (DC02 handled auth)
Update DC01's DNS entry in pfSense DHCP to point to 10.10.10.10 and 10.10.10.11 as DNS servers

13.3 OpenEMR — Full Electronic Medical Record System

[LEARN] LEARN WHY / APPRENDRE POURQUOI

OpenEMR is a complete, HIPAA-compliant Electronic Medical Record system used in real clinics worldwide.
It manages patient demographics, appointments, billing, prescriptions, lab results, and clinical notes.
Installing this transforms your lab from a 'network demo' into a 'clinical IT system demo'.
OpenEMR est un système de dossiers médicaux électroniques complet utilisé dans de vraies cliniques dans le monde entier.

Install on APP01 (10.10.20.20) — SSH in and run:

```
sudo apt install -y apache2 mysql-server php php-mysql php-gd php-curl php-xml php-zip
php-mbstring php-soap
cd /var/www/html
sudo wget https://github.com/openemr/openemr/archive/refs/heads/master.zip -O
openemr.zip
sudo unzip openemr.zip
sudo mv openemr-master openemr
sudo chown -R www-data:www-data /var/www/html/openemr
# Access setup wizard: http://10.10.20.20/openemr
# Database: hospital_db, user: hospital_app, pass: H0sp1tal@2024
```

13.4 Additional Workstations — Scaling the Domain

Add more workstations to demonstrate that the infrastructure scales automatically:

New VM	VLAN	IP	Domain User	Purpose
DOCTOR02	VLAN 30	DHCP: 10.10.30.101	dr.smith@hospital.fr	Second doctor workstation
NURSE01	VLAN 30	DHCP: 10.10.30.102	nurse.claire@hospital.fr	Nurse station
LAB01 (Ubuntu)	VLAN 20	Static: 10.10.20.22	N/A — server	Lab results processing server

New VM	VLAN	IP	Domain User	Purpose
RECEPTION01	VLAN 30	DHCP: 10.10.30.103	m.reception2@hospital.fr	Reception desk

13.5 Suricata IDS/IPS — Intrusion Detection

[LEARN] LEARN WHY / APPRENDRE POURQUOI

Suricata is an Intrusion Detection/Prevention System. It reads every packet passing through pfSense.
IDS (Detection) mode: it alerts on suspicious traffic but lets it through.
IPS (Prevention) mode: it actively BLOCKS traffic matching known attack signatures.
Suricata lit chaque paquet passant par pfSense et alerte/bloque le trafic suspect.

135. pfSense: System → Package Manager → Available Packages → search 'suricata' → Install
136. Services → Suricata → add interface for each VLAN (MGMT, CLINICAL, ADMIN, GUEST)
137. Global Settings tab: subscribe to ET (Emerging Threats) free ruleset → click Update Rules
138. Start with IDS mode → check alerts for 1 week → switch to IPS after tuning false positives
139. Simulate attack: from GUEST01, run: nmap 10.10.20.20 → Suricata detects and alerts in logs

13.6 OpenVPN — Remote Doctor Access

[LEARN] LEARN WHY / APPRENDRE POURQUOI

OpenVPN creates an encrypted tunnel from a doctor's laptop at home to the hospital network.
After VPN connect, the remote laptop is virtually inside the hospital's VLAN — full secure access.
This is standard technology in every healthcare organization offering on-call or remote telemedicine.

140. pfSense: VPN → OpenVPN → Wizards → Local User Access → Next
141. Protocol: UDP on IPv4, Interface: WAN, Tunnel Network: 172.16.0.0/24
142. Local Network: 10.10.0.0/16 (covers all VLANs)
143. Create VPN user: System → User Manager → Add user: vpn.doctor / VPN@2024!
144. VPN → OpenVPN → Client Export → download .ovpn config for the user
145. On your Windows host: install OpenVPN client → import .ovpn → Connect → you are inside the hospital network

13.7 Grafana — Visual Monitoring Dashboards

Install Grafana on APP01 alongside Zabbix for beautiful dashboards:

```
sudo apt install -y apt-transport-https software-properties-common
wget -q -O - https://packages.grafana.com/gpg.key | sudo apt-key add -
echo 'deb https://packages.grafana.com/oss/deb stable main' | sudo tee
/etc/apt/sources.list.d/grafana.list
sudo apt update && sudo apt install -y grafana
sudo systemctl enable grafana-server && sudo systemctl start grafana-server
# Access: http://10.10.20.20:3000 — admin/admin
```

```
# Add Zabbix data source: Configuration → Data Sources → Add → Zabbix
# Create custom 'Hospital Network Health' dashboard with VLAN traffic, server metrics
```

13.8 Course Development Guide / Guide de Développement de Cours

[LEARN] LEARN WHY / APPRENDRE POURQUOI

This section maps the project phases to IT course modules for educators.
Cette section associe les phases du projet aux modules de cours IT pour les formateurs.

Course Module	POC Section	Key Concepts Taught	Duration	Certification Aligned
Module 1: Virtualization	Section 3	VMs, hypervisors, VT-x, snapshots	3 hours	CompTIA A+, VCP
Module 2: Network Fundamentals	Sections 1, 4.1-4.3	IP addressing, subnets, gateways, NAT	4 hours	CompTIA Network+
Module 3: Firewall & Security	Sections 4.5-4.9, 10.1	VLANs, firewall rules, least privilege	5 hours	CompTIA Security+
Module 4: Linux Server Admin	Sections 6, 7	Ubuntu, Netplan, systemctl, SSH, LAMP	6 hours	LPIC-1, Linux+
Module 5: Windows Server / AD	Section 5	ADDS, DNS, GPO, domain join, RBAC	6 hours	MCSA, Server+
Module 6: Healthcare IT	Sections 7, 9	PACS, DICOM, HL7, HIPAA/HDS	4 hours	CPHIMS, CHDA
Module 7: Monitoring & ITSM	Section 9	Zabbix, alerting, SNMP, performance	3 hours	ITIL v4
Module 8: Backup & DR	Section 12	RTO, RPO, snapshot strategy	2 hours	BCCP

14. Troubleshooting Guide / Guide de Dépannage

ℹ INFO / NOTE

Use this section whenever something is not working. Issues are grouped by symptom.
Rule: always check the most likely cause first — 90% of VirtualBox VLAN problems are Promiscuous Mode.

14.1 Connectivity Issues

Symptom / Symptôme	Most Likely Cause	Step-by-Step Fix
VM can only ping itself, not gateway	Promiscuous Mode not 'Allow All'	1) Shut down ALL VMs 2) Each VM → Settings → Network → each adapter → Advanced → Promiscuous Mode → Allow All 3) Boot in order: FW01 → DC01 → APP01 → MED01
pfSense cannot ping any VM	Promiscuous Mode on pfSense Adapter 2	FW01 → Settings → Network → Adapter 2 → Advanced → Allow All → Reboot FW01
Inter-VLAN ping fails (e.g. APP01 can't reach DC01)	Firewall rules blocking traffic	pfSense: Firewall → Rules → CLINICAL_VLAN20 → add Allow Any rule temporarily for testing
VLAN interface shows 'down' in pfSense	Interface not enabled	Interfaces → [VLAN name] → check 'Enable Interface' → Save → Apply Changes
VMs cannot get DHCP IP	DHCP not enabled on VLAN in pfSense	Services → DHCP Server → click VLAN tab → Enable → set range → Save
pfSense web UI unreachable	Wrong IP or interface assigned wrong	FW01 console → Option 2 → set LAN IP to 192.168.1.1 → try from DC01 again

14.2 Windows / Domain Issues

Symptom	Cause	Fix
Cannot join domain hospital.fr	DNS not pointing to DC01	CLIENT01: set DNS to 10.10.10.10 (DC01) → retry domain join
Domain join authentication fails	DC01 unreachable or wrong credentials	ping 10.10.10.10 from CLIENT01 first — must succeed. Credentials: HOSPITAL\Administrator
nslookup hospital.fr fails	DC01 DNS not running or wrong DNS configured	On DC01: Services → DNS → ensure started. Client DNS must be 10.10.10.10
GPO not applying	Replication delay or wrong OU	CLIENT01 CMD: gpupdate /force → gpresult /r to see applied policies
ipconfig shows APIPA (169.254.x.x)	DHCP not responding or static IP misconfigured	Check adapter type (must be Internal Network: LabNet). Check pfSense DHCP for that VLAN

14.3 Ubuntu / Linux Issues

Symptom	Cause	Fix
netplan apply fails with error	YAML indentation error — tabs used instead of spaces	Edit with: nano /etc/netplan/... — ensure 2 spaces per indent, ZERO tabs
Wrong interface name in netplan	Interface is not enp0s3	Run: ip a — note actual name (might be ens3, eth0, etc.) — update netplan accordingly
apache2 not starting	Port 80 in use or config error	sudo systemctl status apache2 → sudo journalctl -xe --unit=apache2 for details
MySQL connection refused	Service not running or wrong password	sudo systemctl start mysql → verify password in /var/www/html/hospital/index.php
Orthanc not accessible on :8042	Orthanc service not running	sudo systemctl status orthanc → sudo systemctl restart orthanc
SSH connection refused	SSH not installed or firewall blocking	sudo apt install openssh-server → sudo ufw allow ssh → sudo systemctl start ssh

14.4 Most Important Diagnostic Commands

[EN] ENGLISH

On Ubuntu (APP01, MED01)

```
ip a                                    # Show all network interfaces and IPs
ip route show                           # Show routing table — check default gateway
ping 10.10.20.1 -c 4                   # Test VLAN gateway
ping 10.10.10.10 -c 4                  # Test DC01 inter-VLAN
nslookup hospital.fr 10.10.10.10        # Test DNS via DC01
sudo systemctl status apache2 mysql     # Check web + DB services
sudo systemctl status orthanc           # Check PACS service
sudo journalctl -xe                     # View recent system errors
cat /etc/netplan/00-installer-config.yaml  # Check current IP config
```

On Windows (DC01, CLIENT01)

```
ipconfig /all                           # Full network configuration
ping 10.10.10.1 -n 4                   # Test gateway
nslookup hospital.fr                    # Test DNS
nltest /sc_query:hospital.fr            # Test DC connection from workstation
gpupdate /force                         # Force Group Policy refresh
net user /domain                        # List all domain users
Get-ADComputer -Filter *                # PowerShell: list all domain computers
```

15. Glossary / Glossaire

ℹ INFO / NOTE

This glossary defines every technical term used in this guide in plain English and French. Reference it whenever you encounter an unfamiliar term.

Term / Terme	Definition (EN)	Définition (FR)
Active Directory (AD)	Microsoft's database of all users, computers, and policies in a domain	Base de données Microsoft de tous les utilisateurs, ordinateurs et politiques d'un domaine
DHCP	Dynamic Host Configuration Protocol — automatically assigns IP addresses	Protocole qui attribue automatiquement des adresses IP aux appareils
DICOM	Digital Imaging and Communications in Medicine — the standard format for medical images	Format standard international pour les images médicales (radiographies, IRM, etc.)
DNS	Domain Name System — translates names (app01.hospital.fr) into IPs (10.10.20.20)	Système qui traduit les noms de domaine en adresses IP
Domain Controller (DC)	The Windows Server that runs Active Directory and authenticates users	Serveur Windows exécutant Active Directory et authentifiant les utilisateurs
Firewall	Network security device that controls what traffic is allowed or blocked	Dispositif de sécurité réseau qui contrôle le trafic autorisé ou bloqué
GPO	Group Policy Object — a set of configuration rules pushed to domain computers automatically	Ensemble de règles de configuration poussées automatiquement aux ordinateurs du domaine
HIPAA	US healthcare data privacy law requiring strict security controls	Loi américaine sur la confidentialité des données de santé exigeant des contrôles de sécurité stricts
HDS	French healthcare data hosting regulation (Hébergeur de Données de Santé)	Réglementation française d'hébergement de données de santé
IDS/IPS	Intrusion Detection/Prevention System — monitors and blocks attack patterns	Système de détection/prévention d'intrusion — surveille et bloque les modèles d'attaque
IP Address	A unique number identifying a device on a network (e.g. 10.10.20.20)	Numéro unique identifiant un appareil sur un réseau (ex: 10.10.20.20)
LAMP	Linux + Apache + MySQL + PHP — the most common web server stack	La combinaison la plus utilisée pour héberger des applications web
LDAP	Lightweight Directory Access Protocol — used to authenticate against Active Directory	Protocole utilisé pour s'authentifier auprès d'Active Directory
NAT	Network Address Translation — allows VMs to share the host's internet connection	Traduction d'adresses réseau — permet aux VMs de partager la connexion internet de l'hôte
Netplan	Ubuntu's YAML-based network configuration system	Système de configuration réseau d'Ubuntu basé sur des fichiers YAML
PACS	Picture Archiving and Communication System — stores and retrieves medical images	Système d'archivage et de communication d'images médicales

Term / Terme	Definition (EN)	Définition (FR)
pfSense	Free BSD-based firewall/router OS used in enterprises and hospitals	OS pare-feu/routeur FreeBSD gratuit utilisé en entreprise et dans les hôpitaux
Promiscuous Mode	VirtualBox setting that allows a VM to receive ALL network traffic, including VLAN tags	Paramètre VirtualBox permettant à une VM de recevoir TOUT le trafic réseau, y compris les tags VLAN
RBAC	Role-Based Access Control — users get access based on their job role	Contrôle d'Accès Basé sur les Rôles — les utilisateurs ont accès selon leur rôle
Snapshot	A saved state of a VM that can be restored instantly — the 'undo' button	État sauvegardé d'une VM qui peut être restauré instantanément — le bouton 'annuler'
SSH	Secure Shell — encrypted remote terminal access to Linux servers	Accès terminal distant chiffré aux serveurs Linux
Static IP	An IP address that never changes — required for servers	Adresse IP qui ne change jamais — obligatoire pour les serveurs
VLAN	Virtual LAN — a logical network partition isolating traffic on one physical switch	Réseau local virtuel — partition logique isolant le trafic sur un commutateur physique
VPN	Virtual Private Network — encrypted tunnel for remote secure access	Réseau Privé Virtuel — tunnel chiffré pour l'accès sécurisé à distance
VirtualBox	Free software from Oracle for running virtual machines	Logiciel gratuit d'Oracle pour exécuter des machines virtuelles
WAN	Wide Area Network — the internet connection side of a router/firewall	Le côté connexion internet d'un routeur/pare-feu
YAML	Indentation-based configuration file format used by Netplan and many tools	Format de fichier de configuration basé sur l'indentation — utilisé par Netplan
Zabbix	Free enterprise monitoring platform for servers, networks, and applications	Plateforme de surveillance d'entreprise gratuite pour serveurs, réseaux et applications

Appendix A — Quick Reference Card / Carte de Référence Rapide

A.1 Complete Service Catalog with Credentials

Service	URL / Access Method	Username	Password	VLAN
pfSense Web UI	https://10.10.10.1 (from DC01 browser)	admin	pfsense	—
pfSense Console	VirtualBox FW01 console directly	—	—	—
Domain Admin	HOSPITAL\Administrator	Administrator	P@ssw0rd	VLAN 10
Dr. Martin	HOSPITAL\dr.martin	dr.martin	Doctor@2024!	VLAN 30 → 20
Nurse Sophie	HOSPITAL\nurse.sophie	nurse.sophie	Nurse@2024!	VLAN 30 → 20
Tech Admin	HOSPITAL\admin.tech	admin.tech	Admin@2024!	VLAN 10
Hospital Portal	http://10.10.20.20/hospital/	N/A	N/A	VLAN 20
Orthanc PACS	http://10.10.20.21:8042	orthanc	orthanc	VLAN 20
Zabbix Monitor	http://10.10.20.20/zabbix	Admin	zabbix	VLAN 20
MySQL hospital DB	APP01 CLI: mysql -u hospital_app -p	hospital_app	H0sp1tal@2024	local
MySQL Zabbix DB	APP01 CLI: mysql -u zabbix -p	zabbix	Z@bb1x2024!	local

⚠ WARNING / AVERTISSEMENT

CHANGE ALL DEFAULT PASSWORDS in any real production deployment. These are lab-only credentials.

A.2 IP Address Summary / Résumé des Adresses IP

VM / Device	IP Address	VLAN	Subnet	Gateway
FW01 — VLAN 10 interface	10.10.10.1	10	10.10.10.0/24	(is the gateway)
FW01 — VLAN 20 interface	10.10.20.1	20	10.10.20.0/24	(is the gateway)
FW01 — VLAN 30 interface	10.10.30.1	30	10.10.30.0/24	(is the gateway)
FW01 — VLAN 40 interface	10.10.40.1	40	10.10.40.0/24	(is the gateway)
DC01 — Domain Controller	10.10.10.10	10	10.10.10.0/24	10.10.10.1
APP01 — Web / Monitoring	10.10.20.20	20	10.10.20.0/24	10.10.20.1

VM / Device	IP Address	VLAN	Subnet	Gateway
MED01 — PACS / Medical	10.10.20.21	20	10.10.20.0/24	10.10.20.1
CLIENT01 — Staff Workstation	10.10.30.30	30	10.10.30.0/24	10.10.30.1
GUEST01 — IoT / Guest	10.10.40.40	40	10.10.40.0/24	10.10.40.1
DC02 (future HA)	10.10.10.11	10	10.10.10.0/24	10.10.10.1

A.3 VM Boot Order & RAM Management

Boot Step	VM	Boot Wait	RAM	Use Headless?
1st — ALWAYS FIRST	FW01 (pfSense)	60 seconds	1 GB	Yes — right-click → Start Headless
2nd	DC01 (Windows Server)	3 minutes	2 GB	Yes — once confirmed working
3rd	APP01 (Ubuntu)	30 seconds	1.5 GB	Yes
4th	MED01 (Ubuntu)	30 seconds	1 GB	Yes
5th — only when testing	CLIENT01 (Win 11)	2 minutes	1.5 GB	No — needs console for demo
Only when needed	GUEST01	1 minute	1 GB	No — needs console for demo

A.4 Phase Completion Checklist / Liste de Contrôle des Phases

Phase	Description	Complete?	Notes
Phase 1	All 6 VMs created, network adapters configured, Promiscuous Mode set	[ ]	Section 3
Phase 2	pfSense installed, interfaces assigned, 4 VLANs created with DHCP	[ ]	Section 4
Phase 3	DC01: Windows Server installed, AD DS, DNS, domain hospital.fr active	[ ]	Section 5
Phase 4	APP01: Ubuntu installed, static IP, LAMP, hospital portal live	[ ]	Section 6
Phase 5	MED01: Ubuntu installed, static IP, Orthanc PACS running	[ ]	Section 7
Phase 6	CLIENT01 & GUEST01: installed, CLIENT01 domain-joined	[ ]	Section 8
Phase 7	Zabbix monitoring deployed, all VMs added as hosts	[ ]	Section 9
Phase 8	Production firewall rules applied, GPOs configured	[ ]	Section 10
Phase 9	All 15 connectivity tests pass, all 6 service checks pass	[ ]	Section 11

Phase	Description	Complete?	Notes
Phase 10	VirtualBox snapshots taken, backup script running	[ ]	Section 12

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